ELECTRICAL—VEHICLE SYSTEMS GROUND LOCATIONS—INTERMITTENT FAULT DIAGNOSTIC TIPS	Article No. 02-12-2

LINCOLN: 2000-2002 LS

Article **00-25-7** is being republished in its entirety to update the model year coverage.

ISSUE

The Lincoln LS electrical system is of a multiplex design. Multiplexed electrical systems rely very heavily on the integrity of the component and circuit grounds to function correctly. Intermittent electrical faults may be caused by a poor ground.

ACTION

When diagnosing any intermittent electrical faults, the grounds for the affected system or component should be inspected for proper connection and integrity. Poor or high resistance grounds can affect the operation of modules, relays and other components, but may not generate trouble codes. Refer to the following Ground Location Chart for ground locations and descriptions of which components/systems are on each specified ground.

GROUND LOCATIONS AND DESCRIPTION

Refer to Figure 1 for Vehicle Ground Location overview.

NOTE

TO PROPERLY INSPECT A GROUND, FIRST COMPLETELY REMOVE THE FASTENER RETAINING THE GROUND TO THE VEHICLE. INSPECT EACH INDIVIDUAL GROUND EYELET FOR DAMAGE TO THE TERMINAL AND WIRE. REPAIR DAMAGE AS NEEDED TO RESTORE THE CONDUCTIVITY OF THE GROUND. THROUGHLY CLEAN, REASSEMBLE AND TORQUE THE GROUND FASTENER TO THE RECOMMENDED VALUE FOUND IN THE APPROPRIATE WORKSHOP MANUAL.

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GROUND LOCATION TABLE	
Ground Location	**Systems Grounded At This Location**
G100 - Located behind the LH headlamp housing (Figure 2)	Brake fluid level switch Windshield wiper motor Hood ajar switch LH driving lamp Thermactor air pump relay (V6 only) Horn
G101 - Located behind the RH headlamp housing (Figure 3)	Air conditioning clutch Powertrain Control Module (PCM) relay Washer reservoir level switch Traction Control System or Interactive Vehicle Dynamics (Advance Trac) module Ignition coil on plug relay Auxiliary coolant pump (V8 only) RH driving lamp Wiper motor
G102 - Located in the RH front wheelwell. **Note:** The wheelwell plastic liner must be removed to access G102 (Figure 4)	Crankshaft position sensor Mass Air Flow (MAF) sensor Powertrain Control Module (PCM)
G103 - Located in the RH front wheelwell. **Note:** The wheel well plastic liner must be removed to access G103 (Figure 5)	Starter motor ground
G201 - Located in the instrument panel on the left radio support bracket (Figure 6)	Radio CD player (for in-glove compartment CD player) Tilt/Telescopic steering column Dual Automatic Temperature Control (DATC) module Power mirror switch Trunk release switch Glove compartment lamp Power front amplifier
G202 - LH A-pillar area, behind kick panel (Figure 7)	Front Electronics Module (FEM) Anti-lock Brake System (ABS) test connector
G203 - LH A-pillar area, behind kick panel (Figure 7)	Front Electronics Module (FEM) for the front passenger window Heated wiper park Auxiliary junction box Driver's Door Module (DDM) Driver door switches Driver door latch Power mirror switch Deck lid release switch Passive Anti-Theft System (PATS)

Ground Location	Systems Grounded At This Location
G204 - RH A-pillar area, behind kick panel (Figure 8)	Climate control blower motor Passenger door latch Passenger side heated mirror Dual Automatic Temperature Control (DATC) module
G300 - Located on the RH side of the rear package tray, under the package tray trim cover. **Note:** The trim cover must be removed to access this ground (Figure 9)	Fuel pump relay Headliner lamps and switches RH rear door latch RH rear window switch Moonroof motor/module (if equipped) Deck lid ajar switch
G301 - Located under the RH front seat, under the carpeting (Figure 10)	RH power seat control switches RH heated seat module (if equipped)
G302 - Located under the LH front seat, under the carpeting (Figure 11)	Cigar lighter Traction Control System de-activation switch Transmission shift lever Heated seat switches Heated seat module LH power seat switches
G324 - Located on the LH side of the rear package tray, under the package tray trim cover. **Note:** The trim cover must be removed to access this ground (Figure 12)	Cellular Phone antenna
G400 - LH rear trunk panel, under the plastic trim cover (Figure 13)	Subwoofer amplifier (if equipped) LH rear door latch LH rear power window switch Cellular phone module (if equipped) Remote Emergency Satellite Cellular Unit (RESCU) module (if equipped)
G401 - RH rear trunk panel, under the plastic trim cover (Figure 14)	Rear Electronics Module (REM) Deck lid release solenoid
G403 - Near the battery inside the spare tire well (Figure 15)	This is the main battery ground for the vehicle

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 00-25-7
WARRANTY STATUS: INFORMATION ONLY

OASIS CODES: 102000, 103000, 104000, 112000, 201100, 201200, 202000, 203100, 203200, 204100, 204200, 205000, 206000, 207000, 208000, 208300, 303000, 501000, 502000, 504000, 601300, 603300, 698298

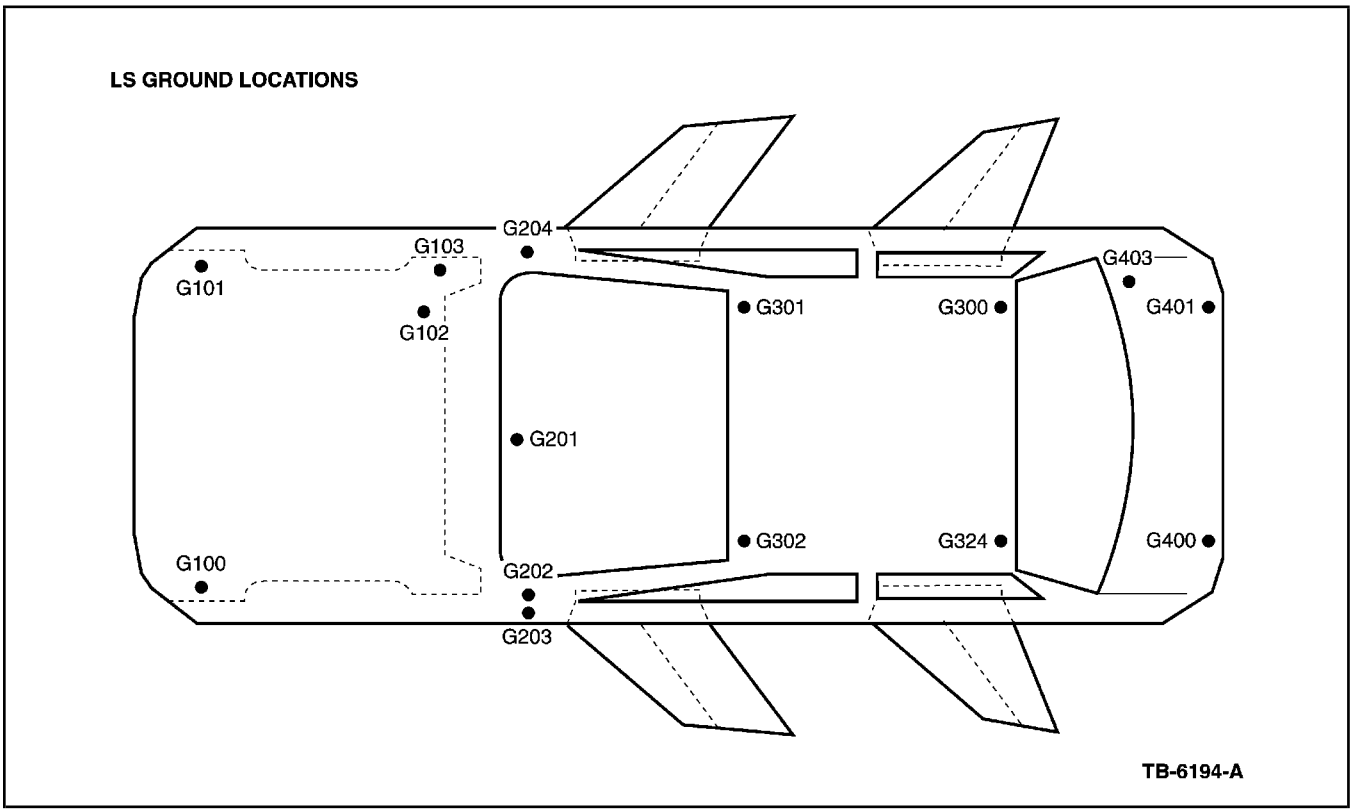
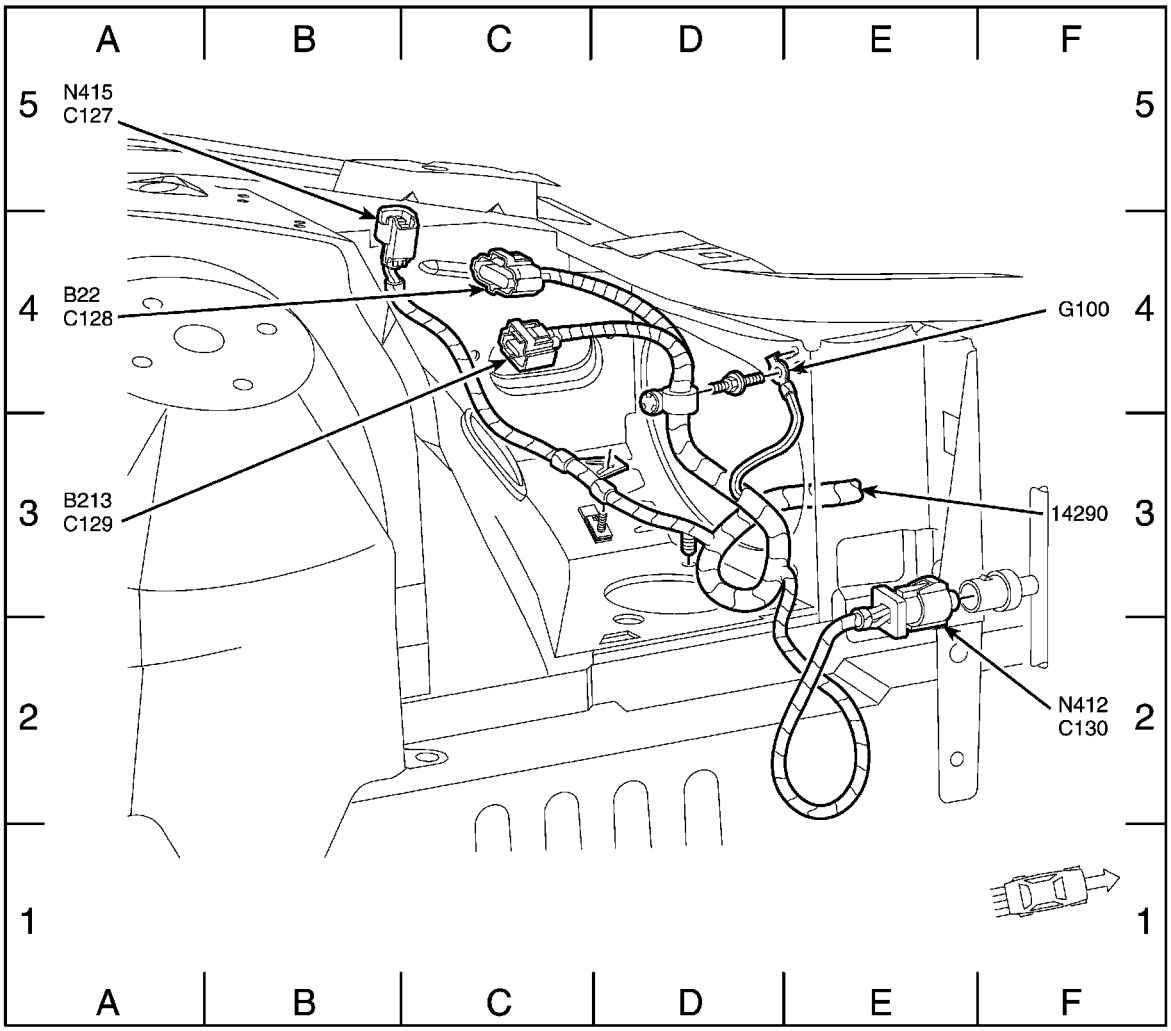


Figure 1 - Article 02-12-2

G100



RADIATOR SUPPORT (LEFT)

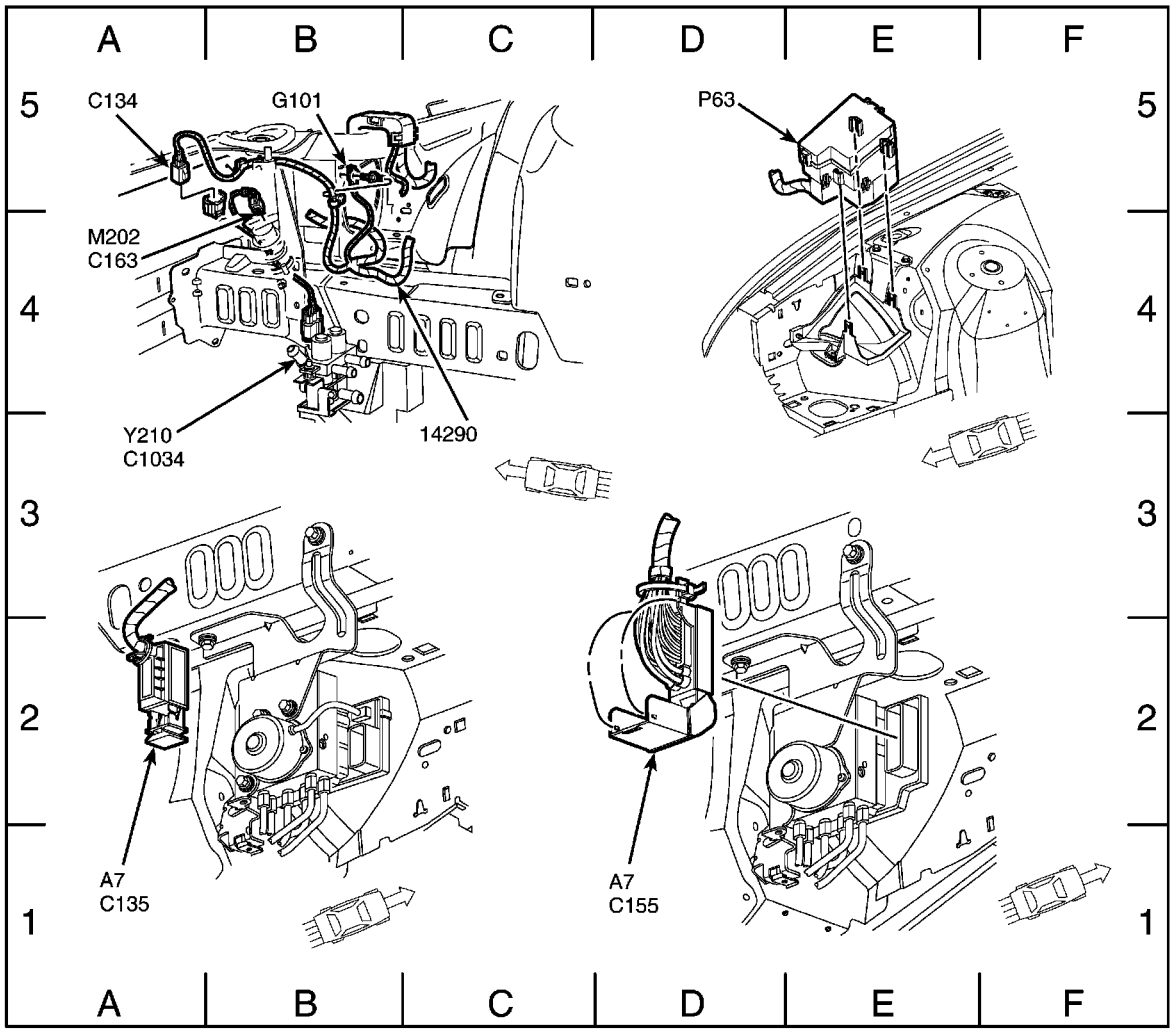
14290		F3
B22	Mass Air Flow (MAF) Sensor	A4
B213	Intake Air Temperature (IAT) Sensor	A3
C127		A5
C128		A4

C129		A3
C130		F2
G100		F4
N412	A/C Clutch Cycling Pressure Switch	F2
N415	Anti-Theft Hood Switch	A5

TB-6195-A

Figure 2 - Article 02-12-2

G101



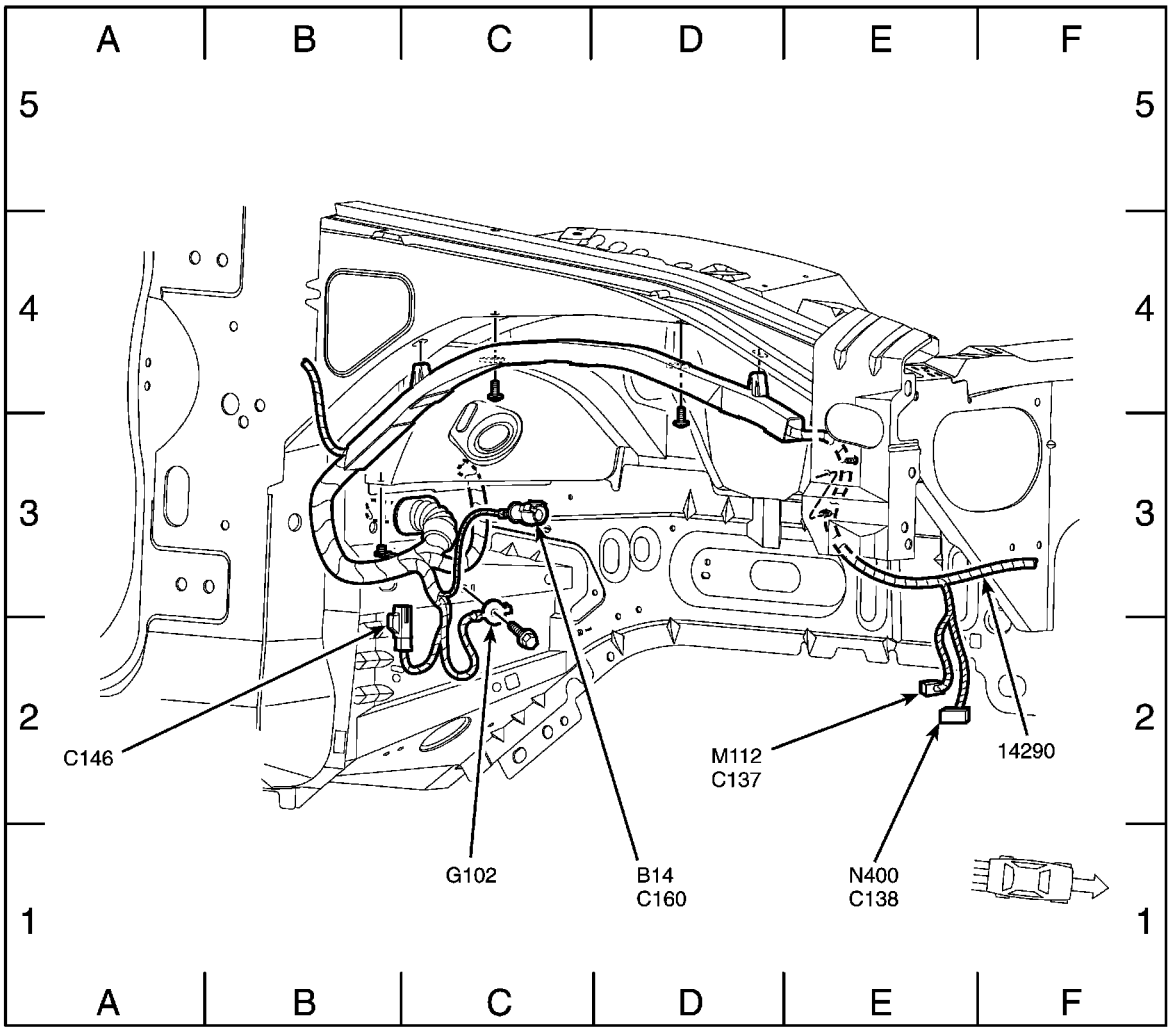
RH ENGINE COMPARTMENT

14290		C3
A7	ABS Control Module	A1
A7	ABS Control Module	D1
C134		A5
C135		A1
C155		D1
C163		A4

C1034		A3
G101		B5
M202	Auxiliary Coolant Pump Motor	A4
P63	Auxiliary Junction Box (AJB), Underhood	D5
Y210	Coolant Control Valve Solenoid	A3

Figure 3 - Article 02-12-2

G102



FENDER (RIGHT FRONT)

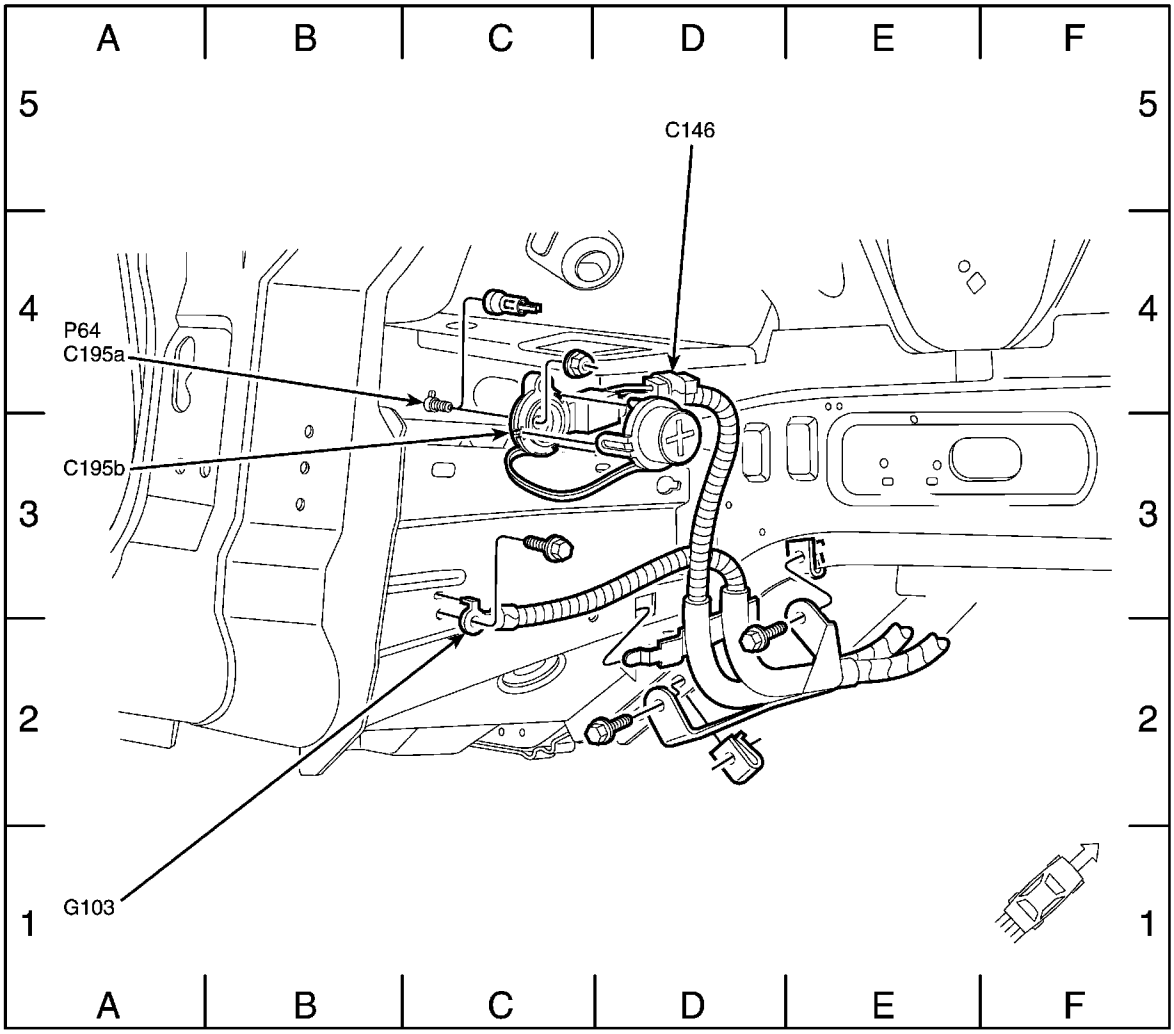
14290		F2
B14	Wheel Speed Sensor, Right Front	D1
C137		D2
C138		E1
C146		A2

C160		D1
G102		C1
M112	Windshield Washer Pump Motor	D2
N400	Washer Fluid Level Switch	E1

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Figure 4 - Article 02-12-2

G103



STARTER SUB-SYSTEM

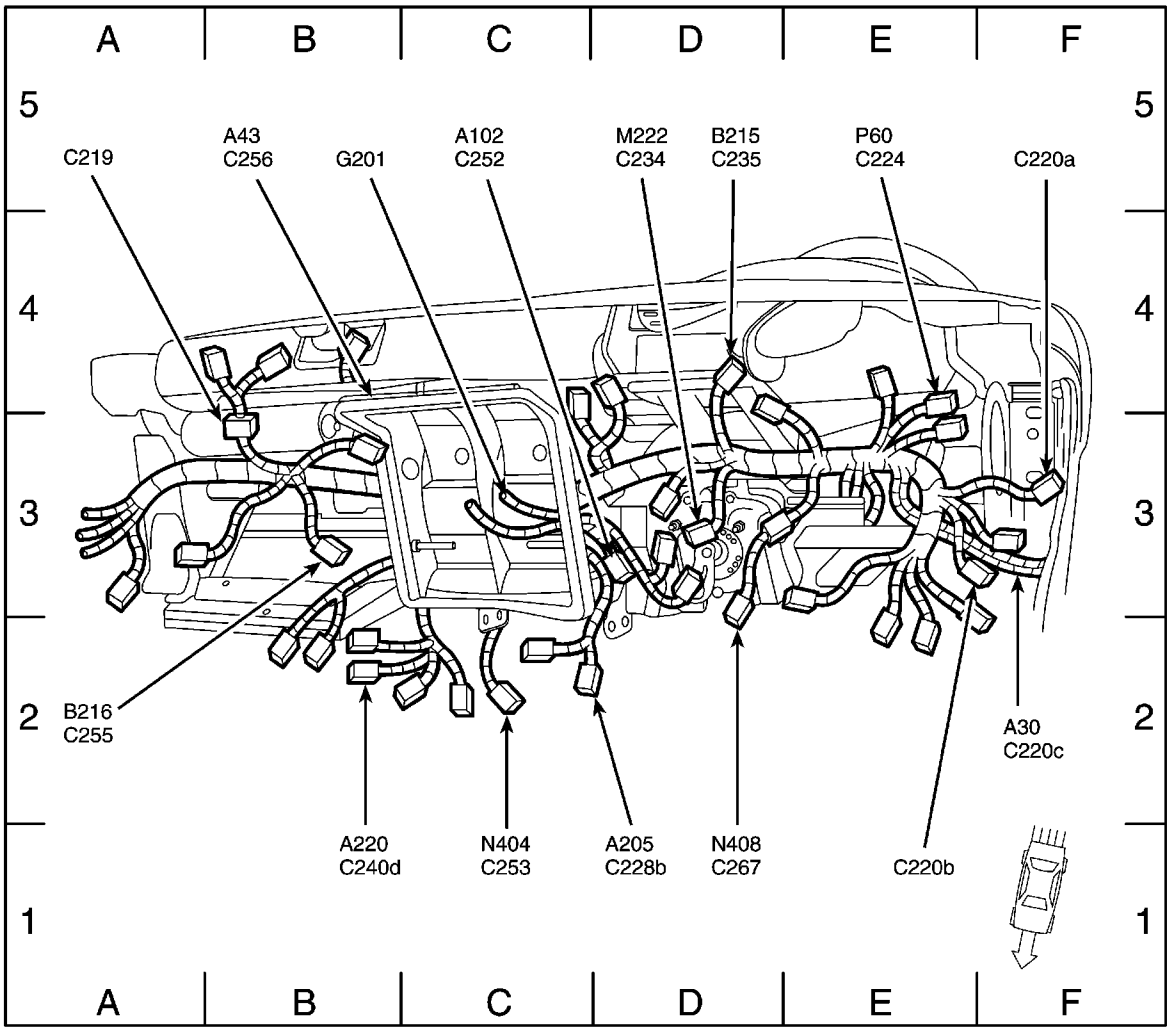
C146		D5
C195a		A4
C195b		A3

G103		A1
P64	Stud Dash Panel	A4

TB-6198-A

Figure 5 - Article 02-12-2

G201



DASH PANEL (REAR VIEW)

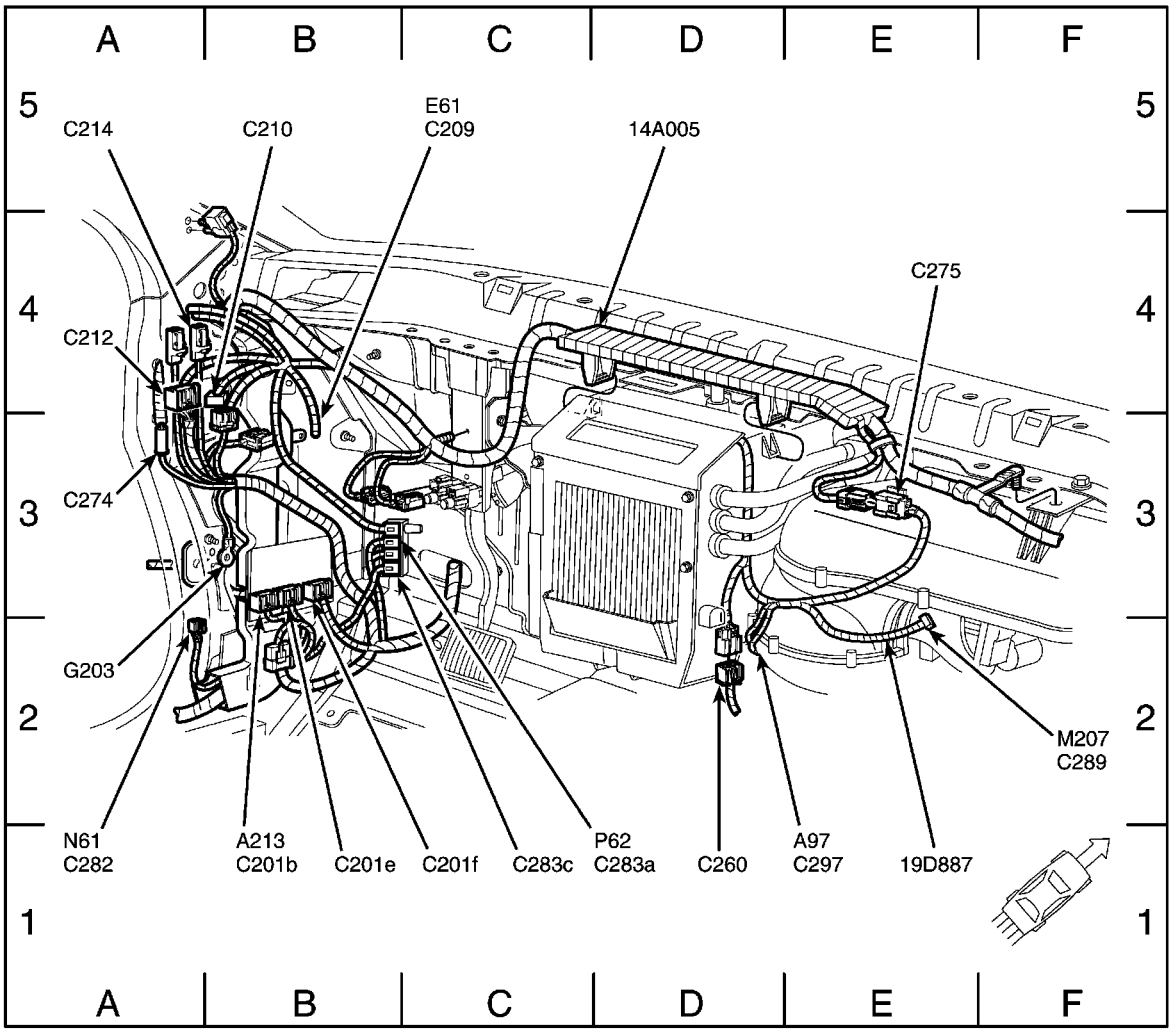
A30	Instrument Cluster	F2
A43	Passenger Air Bag Module	B5
A102	Passive Anti-Theft Transceiver Module	C5
A205	Electronic Automatic Temperature Control (EATC) Module	D1
A220	Audio Unit	B1
B215	Discharge Temperature Sensor, Left	D5
B216	Discharge Temperature Senosr, Right	A2
C219		A5
C220a		F5
C220b		E1
C220c		F2
C224		E5
C228b		D1

C234		D5
C235		D5
C240d		B1
C252		C5
C253		C1
C255		A2
C256		B5
C267		D1
G201		B5
M222	Floor Mode Actuator	D5
N404	Message Center Switch	C1
N408	Steering Column Lock	D1
P60	Joint Connector 3	E5

TB-6199-A

Figure 6 - Article 02-12-2

G202 AND G203



DASH PANEL (LH FRONT VIEW)

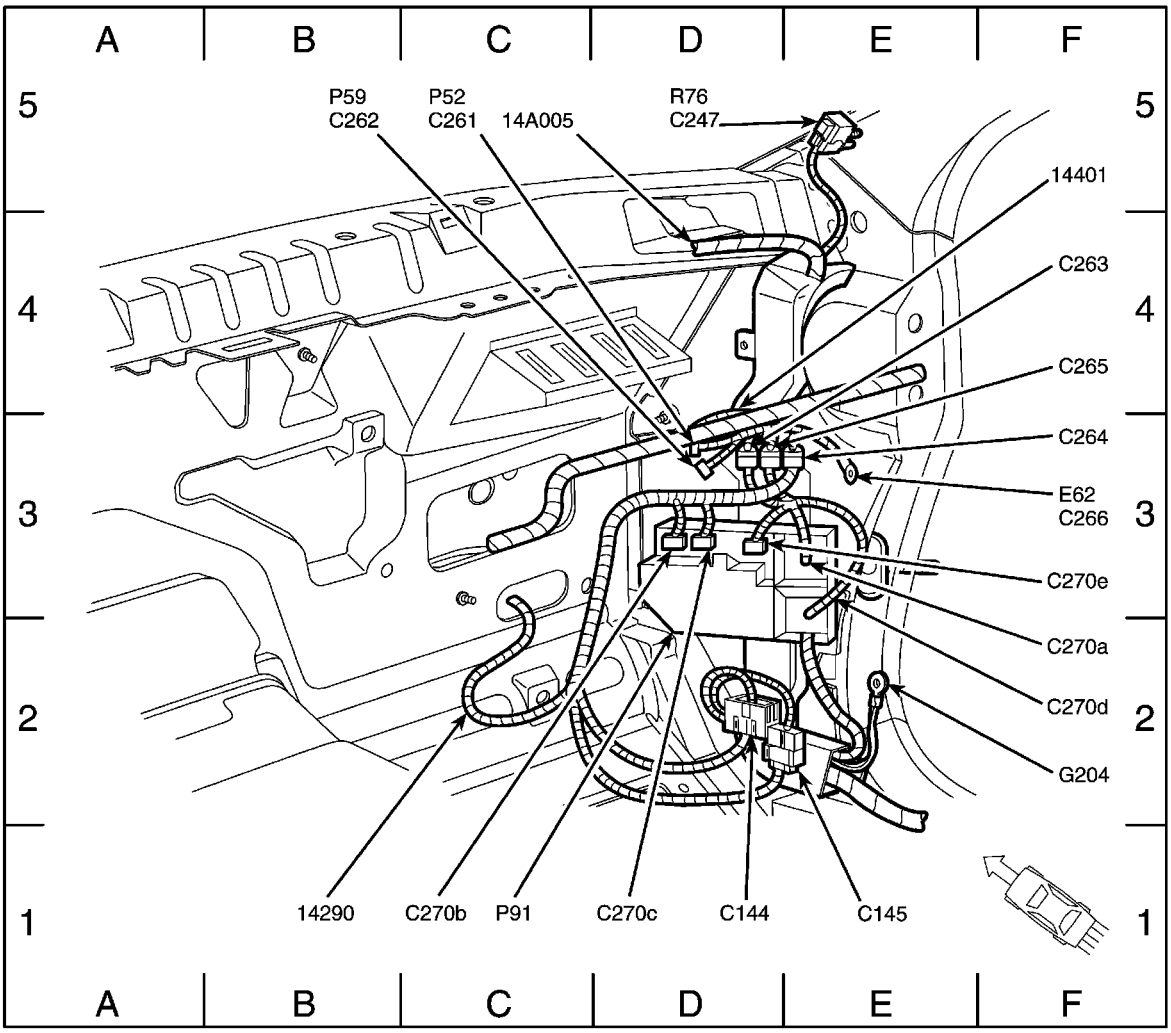
14A005		D5
19D887		E1
A97	Heater Blower Control Module	E1
A213	Front Electronic Module (FEM)	B1
C201b		B1
C201e		B1
C201f		C1
C209		C5
C210		B5
C212		A4
C214		A5
C260		D1

C274		A3
C275		E4
C282		A1
C283a		D1
C283c		C1
C289		F2
C297		E1
E61	Footwell Lamp, Left Front	C5
G203		A2
M207	Fresh/Recirculation Door Actuator	F2
N61	Inertia Fuel Shutoff (IFS) Switch	A1
P62	Auxiliary Junction Box (AJB), Interior	D1

TB-6200-A

Figure 7 - Article 02-12-2

G204



DASH PANEL (RH FRONT VIEW)

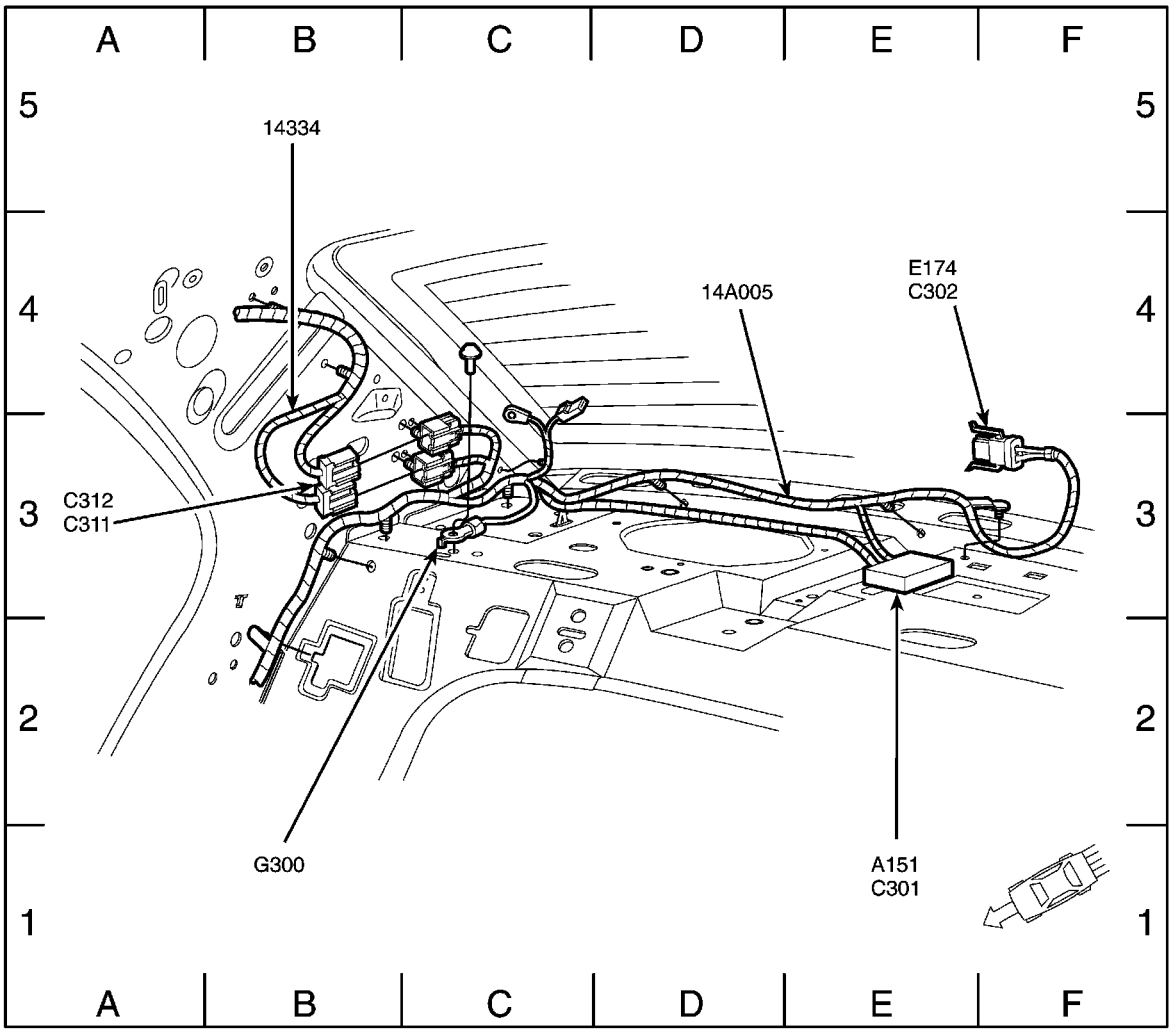
14A005		C5
14290		B1
14401		F5
C144		D1
C145		E1
C247		D5
C261		C5
C262		B5
C263		F4
C264		F3
C265		F4
C266		F3

C270a		F2
C270b		C1
C270c		D1
C270d		F2
C270e		F3
E62	Footwell Lamp, Right Front	F3
G204		F2
P52	Joint Connector 1	C5
P59	Joint Connector 2	B5
P91	Central Junction Box (CJB)	C1
R76	Window Heater, Right	D5

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Figure 8 - Article 02-12-2

G300



PASSENGER TRAY (RIGHT REAR)

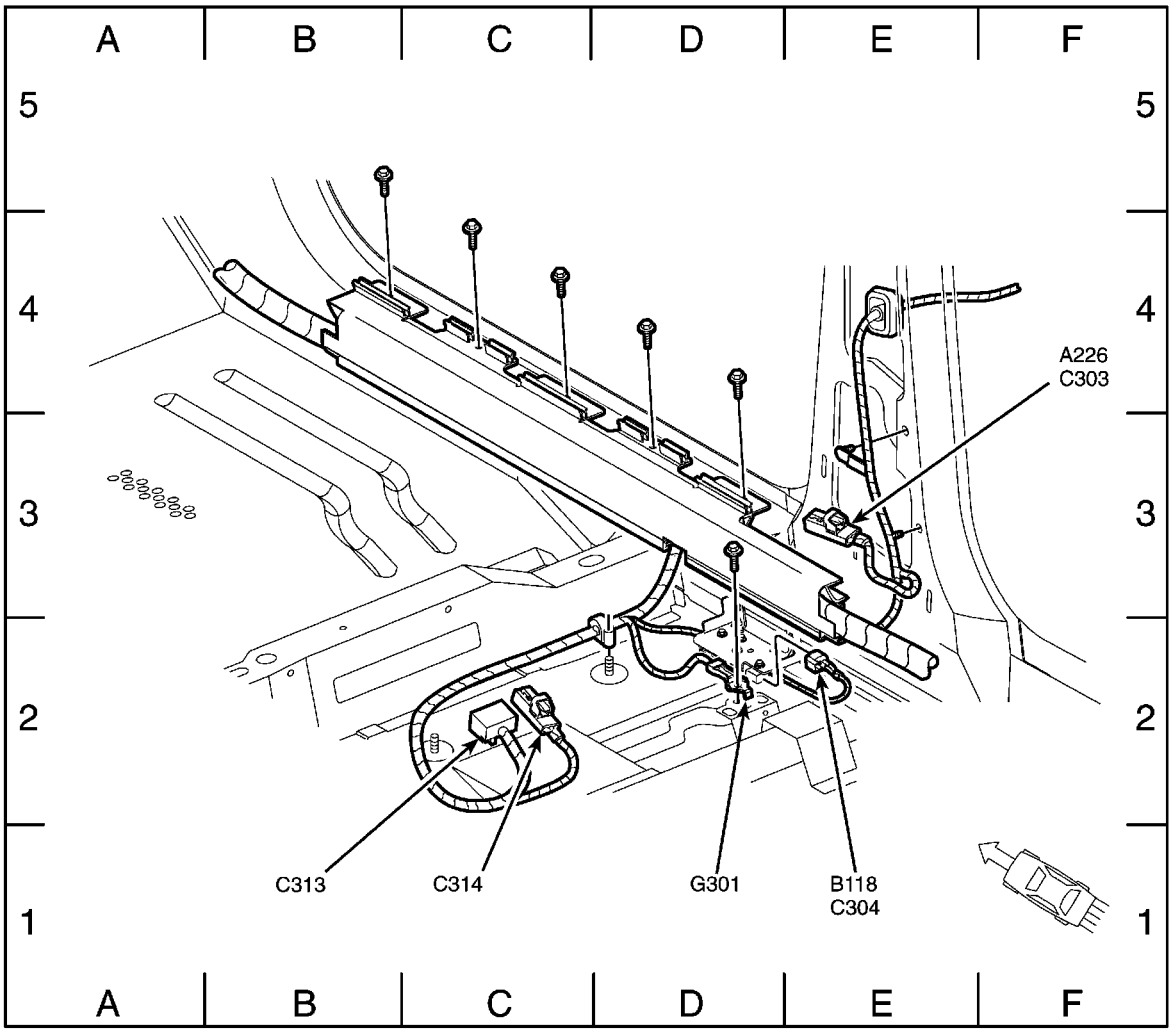
14A005		D4
14334		B5
A151	Antenna	E1
C301		E1
C302		E4

C311		A3
C312		A3
E174	High Mounted Stoplamp	E4
G300		B1

TB-6202-A

Figure 9 - Article 02-12-2

G301



VEHICLE FLOOR (RIGHT FRONT SIDE)

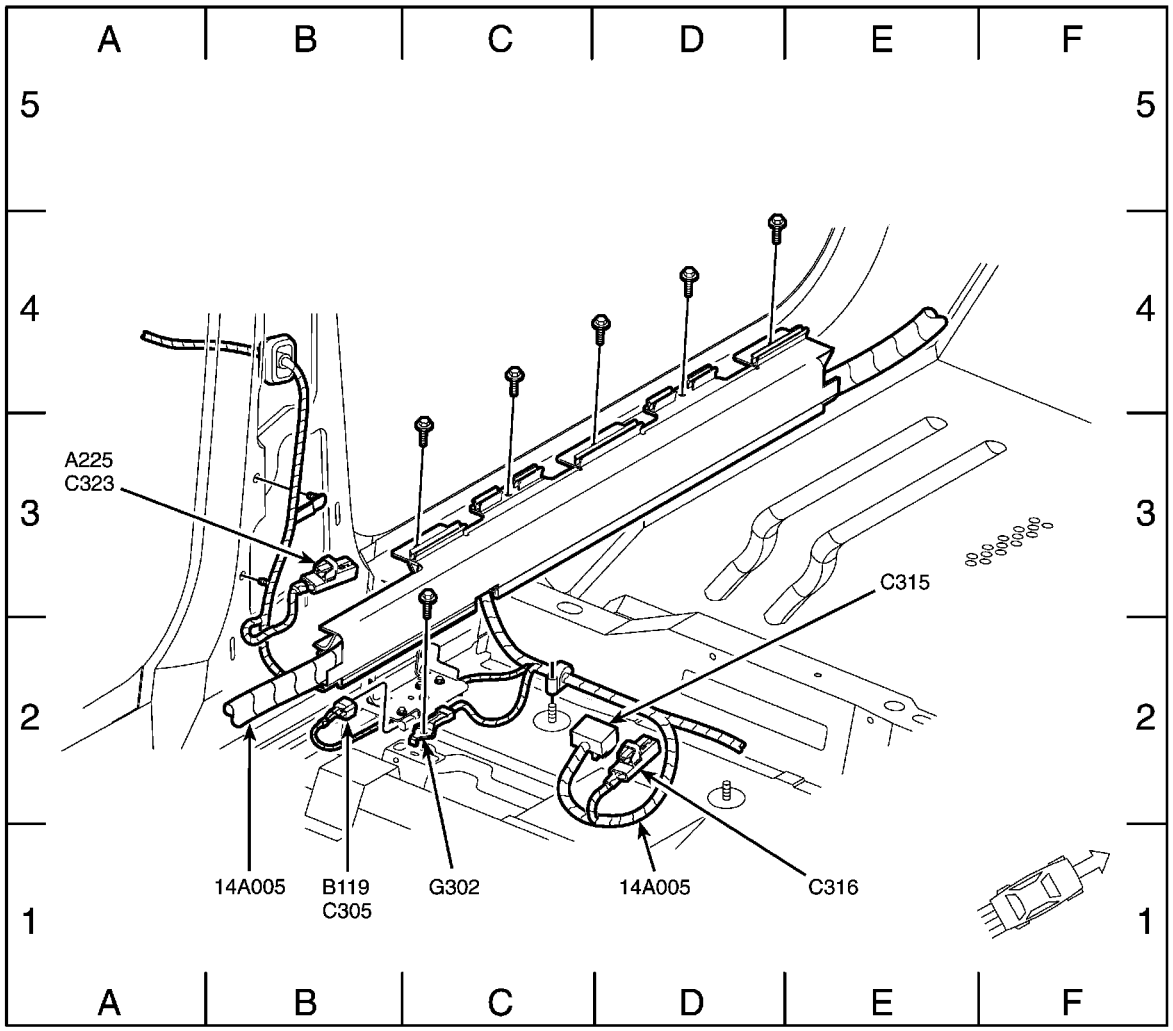
A226	Passenger Safety Belt Retractor Pretensioner	F4
B118	Side Air Bag Sensor, Right	E1
C303		F4

C304		E1
C313		B1
C314		C1
G301		D1

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Figure 10 - Article 02-12-2

G302



VEHICLE FLOOR (LEFT FRONT SIDE)

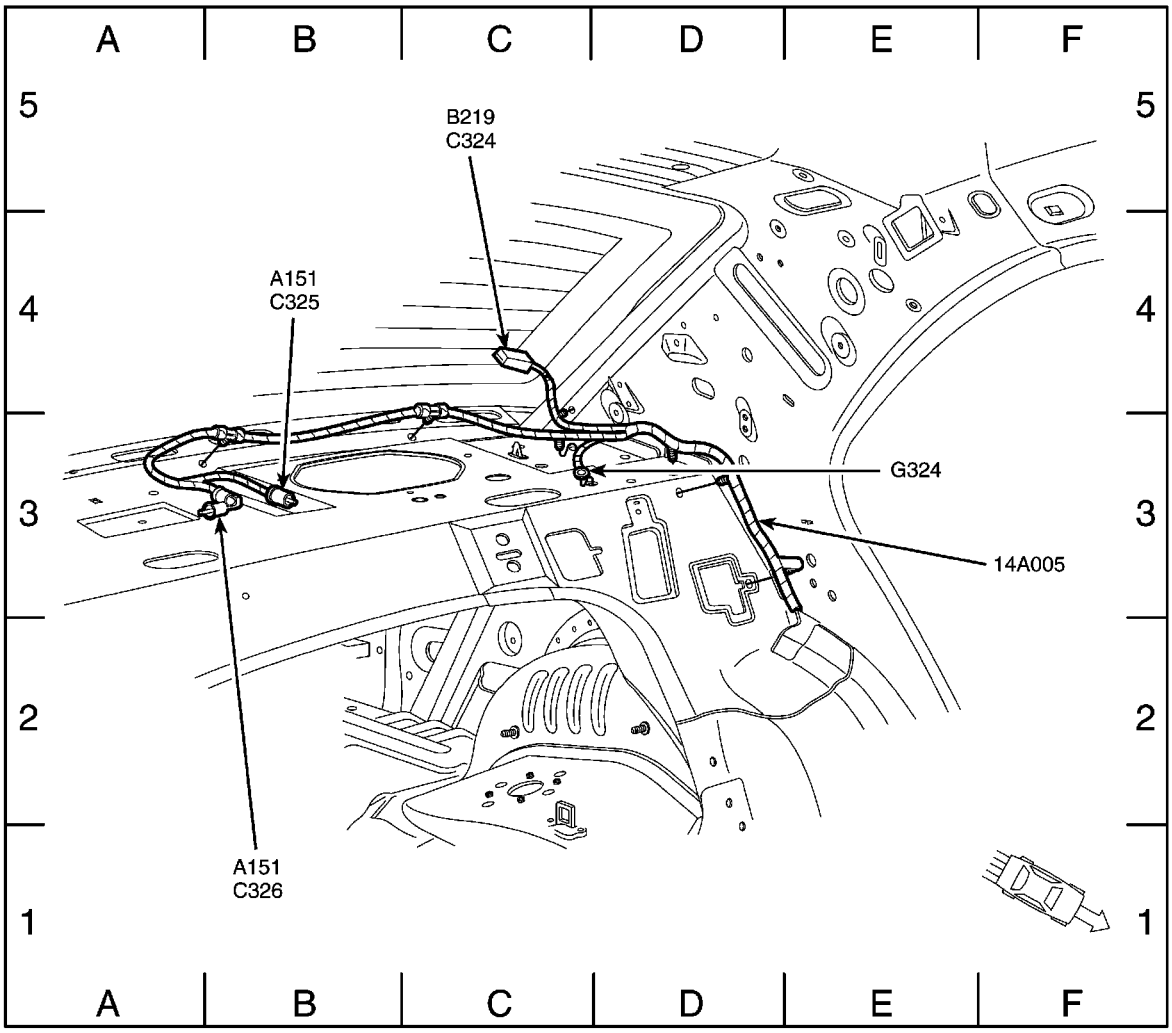
14A005		B1
14A005		D1
A225	Driver Safety Belt Retractor Pretensioner	A3
B119	Side Air Bag Sensor, Left	B1

C305		B1
C315		E3
C316		E1
C323		A3
G302		C1

TB-6204-A

Figure 11 - Article 02-12-2

G324



PACKAGE TRAY (LEFT REAR)

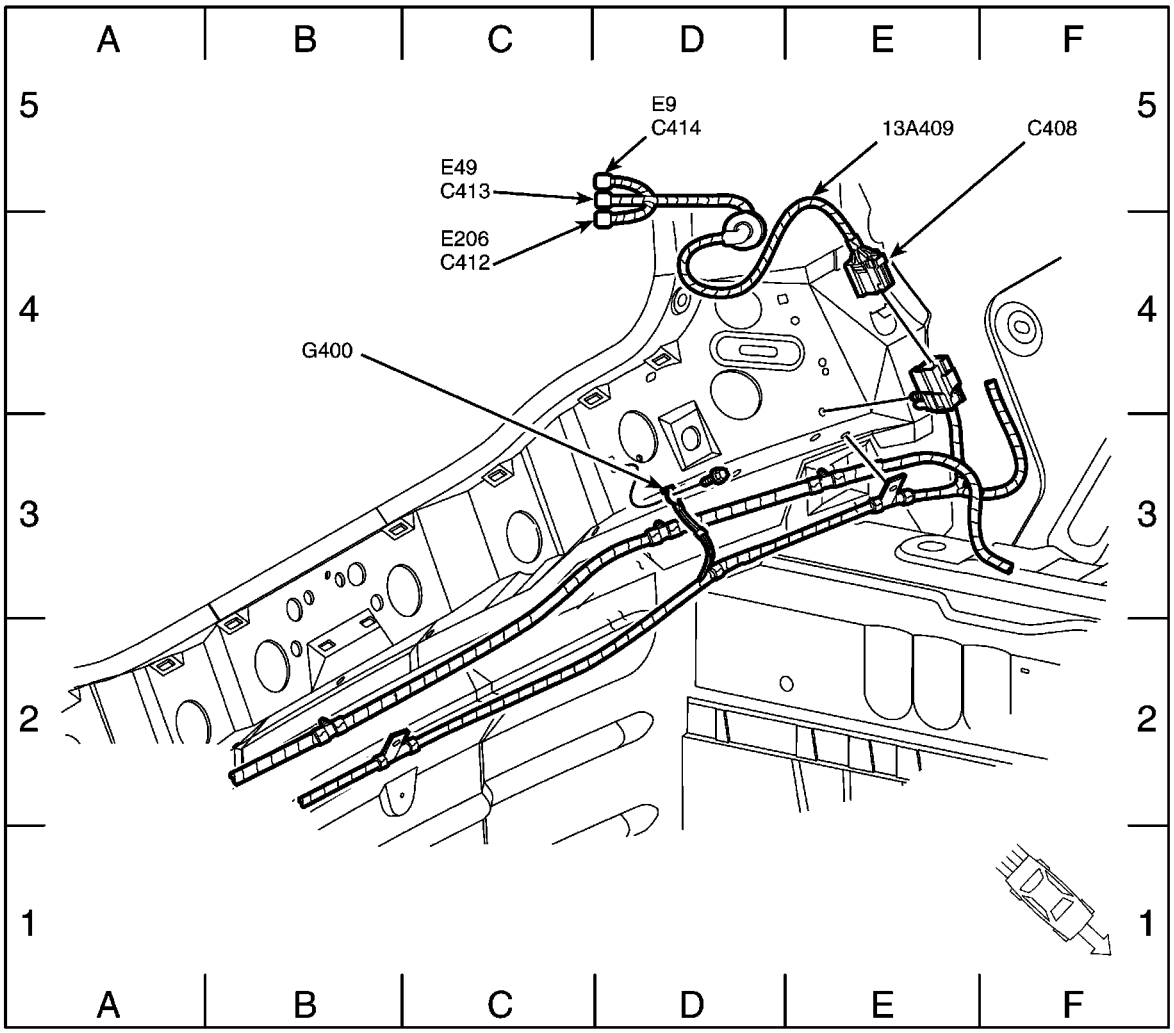
14A005		F3
A151	Antenna	B4
A151	Antenna	B1
B219	Cellular Phone Antenna	C5

C324		C5
C325		B4
C326		B1
G324		E3

TB-6205-A

Figure 12 - Article 02-12-2

G400



LUGGAGE COMPARTMENT (LH SIDE REAR)

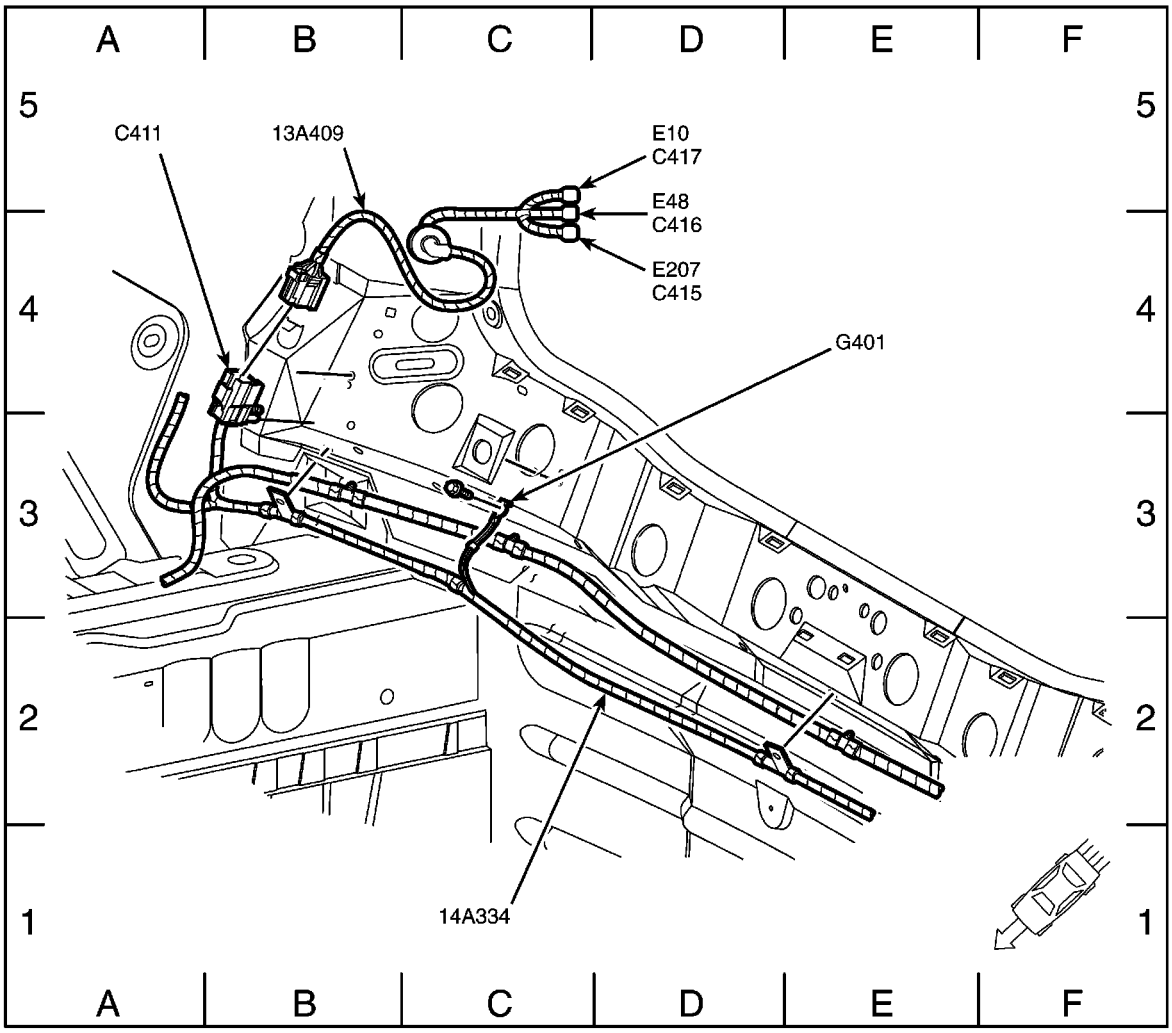
13A409		E5
C408		F5
C412		C4
C413		C5
C414		D5

E9	Lamp Assembly, Left Rear	D5
E49	Turn Lamp, Left Rear	C5
E206	Tail Lamp, Left	C4
G400		B4

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Figure 13 - Article 02-12-2

G401



LUGGAGE COMPARTMENT (RH SIDE REAR)

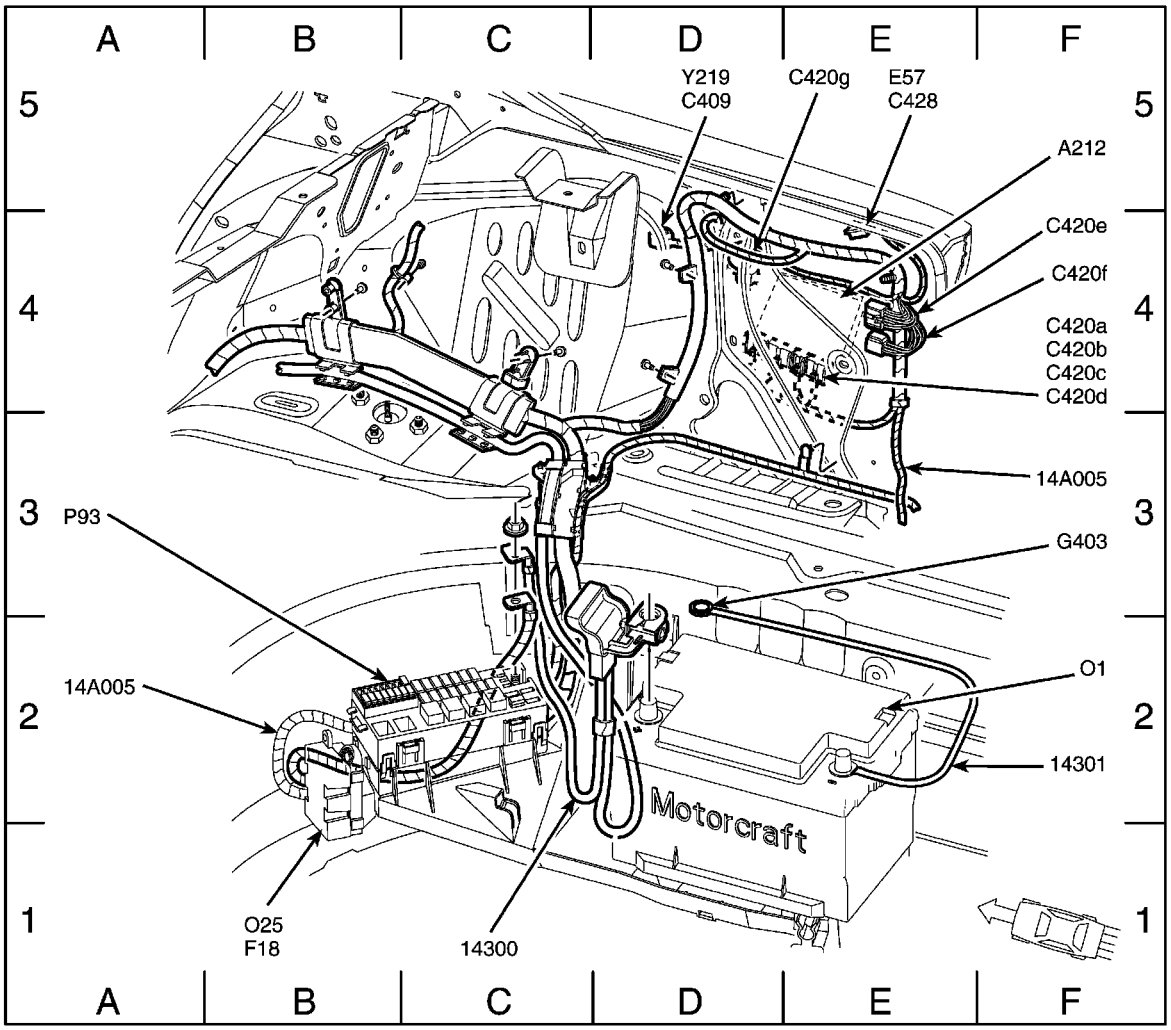
13A409		B5
14A334		C1
C411		A5
C415		D4
C416		D4

C417		D5
E10	Lamp Assembly, Right Rear	D5
E48	Turn Lamp, Right Rear	D4
E207	Tail Lamp, Right	D4
G4001		E4

TB-6207-A

Figure 14 - Article 02-12-2

G403



LUGGAGE COMPARTMENT (RH SIDE FRONT)

14A005		A2
14A005		F3
14300		C1
14301		F2
A212	Rear Electronic Module (REM)	F5
C409		D5
C420a		F4
C420b		F4
C420c		F4
C420d		F4
C420e		F4

C420f		F4
C420g		E5
C428		E5
E57	Luggage Compartment Lamp	E5
F18		B1
G403		F3
O1	Battery	F2
O25	Mega Fuse	B1
P93	Battery Junction Box (BJB)	A3
Y219	Fuel Door Release Solenoid	D5

TB-6209-A

Figure 15 - Article 02-12-2